

AFRICA MEDICAL COLLEGE

Final Examination in Histology and Embryology for Year I Medical Students

Name: _____

Time allowed: 3:00 hours.

Date: Feb 22, 2012

I. HISTOLOGY

MCQ: For each of the following Multiple Choice Questions choose the one BEST appropriate answer and mark 'X' in the appropriate box located against corresponding question number and letter in the separate answer **sheet** provided.

_____1. Which of the following organelles is the site of steroid hormone synthesis?

- a) Golgi complex b) mitochondria c) lysosomes d) RER e) SER

_____2. With regard to the **cellular organelles**, which statement is **true**?

- a) Clathrin-coated vesicles are associated with constitutive protein transport.
b) Lysosomal enzymes have optimal activity at basic pH to synthesize rRNA.
c) Golgi apparatus possess heterochromatin which is transcriptionally active.
d) The nucleolus is a well-defined, membrane-bounded acidophilic nuclear inclusion.
e) Peroxisomes are formed by fusion of phagocytic vacuoles with late endosomes and lysosomes.

_____3. Concerning the morphology of a cell, select the correct statement.

- a) Ribosomes are membranebounded organelles and are involved in the breakdown of hydrogen peroxide.
b) G₀ phase is the longest and most variable cell-cycle phase in continuously dividing cells.
c) Ion-transporting cells contain abundant basal mitochondria between infoldings of the basal plasma membrane
d) Microtubules are more stable than microfilaments.
e) Centrioles are associated with membrane activities such as endocytosis, exocytosis, and cell migratory activities.

_____4. With regard to the epithelial tissue, select the incorrect statement.

- a) The lining of the bladder is transitional epithelium.
b) holocrine secretion is the typical mode of secretion of DNES cells.
c) An example of epithelioid tissue is the parenchyma of the adrenal gland.
d) Merocrine secretion is characterized by discharge of a secretion product with no loss of cytoplasm or membrane.
e) Goblet cells are associated with simple columnar epithelium of the gastrointestinal tract.

_____5. With regard to the **epithelial tissue**, which statement is **true**?

- a) Arteriovenous anastomosis provides nutrition to the highly cohesive rows of cells.
b) Desmosomes permit the passage of ions between adjacent cells.
c) Microvilli contain sheaf of microtubules with a 9+2 axoneme pattern.

- d) The major constituents of epithelia is abundant extracellular matrix with amorphous ground substance.
e) In pseudostratified columnar epithelium all cells touch the basal lamina.

———6. One of the following structures is lined by **stratified columnar epithelium**.

- a) Urinary bladder. b) Conjunctiva. c) Esophagus. d) Vagina. e) Trachea.

———7. The distance between two **Z-lines in skeletal muscle** is equivalent to the length of the:

- a. A band b. H band c. I band d. M line e. sarcomere

———8. The sense organ most closely associated with **stretch reflexes** (e.g., knee jerk) is the:

- a. Meissner's corpuscle. B. Pacinian corpuscle. C. muscle spindle. D. Golgi tendon organ.
E. free nerve ending.

———9. Which of the following is **not** a **function of the kidney**?

- a. Erythropoietin production b. Vitamin D modification c. Acid base balance
d. Aldosterone production e. Renin production

———10. **What vessels arise from the efferent arterioles?**

- a. Afferent arteriole b. Efferent arteriole c. Capillary tuft d. Peritubular capillaries e. Vasa recta**

———11. Select the **correct statement** pertaining the **morphology of the kidney**.

- a. The distal convoluted tubules exhibit brush borders and apical canaliculi.
b. Macula densa cells are located in the urinary pole of the nephrons.
c. Renin is secreted from the visceral layer of Bowman's capsule referred to as area cribrosa.
d. Interstitial cells in the renal medulla secrete erythropoietin through filtration slits.
e. The renal pyramid and its associated cortex is referred to as the renal lobe.

———12. In the **urinary system**, which statement is **false**?

- a. The ureter is lined by transitional epithelium and it does not contain lymphatics.
b. Polyploidy and binucleated cells are common among cells of the transitional epithelium.
c. Epithelial cells lining the thin limb of the loop of Henle: are simple squamous.
d. Mesangial cells are modified smooth muscle cells responsible for the secretion of rennin
e. Aldosterone acts on the distal convoluted duct.

———13. Which of the following hormones is carried to its target(s) by the **hypophyseal portal system**?

- a) ADH b) ACTH c) FSH d) TRH e) TSH

———14. Which statement is **true** with regard to the **endocrine system**?

- a) The chromophobes of pars distalis of adenohypophysis maintain favorable fluid and electrolyte environment for the chromophil secretory cells.
b) The secretory activity of the epiphysis cerebri {pineal body} is stimulated by darkness and inhibited by light.
c) Calcitonin, secreted by the parathyroid gland, stimulates calcium resorption in the bones.
d) The cells of the zona fasciculata of the suprarenal gland are called chromaffin cells because the granules of these cells can be stained with potassium bichromate.
e) The basophilic oxyphil cells are the predominant cell type in the parathyroid gland.

———15. The **neurohypophysis**,

- a) contains chromophils.
- b) includes the pars intermedia.
- c) secretes ACTH.
- d) contains axons of hypothalamic neurons.
- e) hormone production is regulated by releasing factors.

———16. **Testis** has the following coverings **except**:

- a. tunica adventitia
- b. tunica vasculosa
- c. tunica albuginea
- d. tunica vaginalis
- e. None

———17. In the **testis**, which statement is **true**?

- a. Interstitial cells (of Leydig) produce testicular fluid.
- b. The seminiferous tubules are lined with a complex stratified epithelium.
- c. The secondary spermatocytes are the largest and long-lived cells of the spermatogenetic lineage.
- d. The tunica vaginalis is thickened on the posterior surface to form the mediastinum testis.
- e. During spermiogenesis, the spermatids quickly undergo the second meiotic division to give spermatozoa.

———18. Which of the following statements is **false** in the **male reproductive system**?

- a. Tunica albuginea is the thickest in epididymis.
- b. The subcutaneous tissue in penis is devoid of fat.
- c. The secretion of seminal vesicle is particularly rich in fructose.
- d. The prostate gland is composed of compound tubuloalveolar glands.
- e. The epithelium lining the ductus epididymis is pseudostratified columnar.

———19. Which of the following is true of the **ampulla** of the **uterine tube** but **not** of its **isthmus**?

- a) Contains large mucosal folds that are extensively branched.
- b) Contains smooth muscle in its walls.
- c) Is covered by serosa derived from the peritoneum.
- d) Contains ciliated epithelial cells.
- e) Contains mucus-secreting peg cells.

———20. Which of the following statements is **true** about the **female reproductive system**?

- a. Mucosa of the uterus is commonly called perimetrium.
- b. Fornix of vagina is lined by stratified squamous epithelium.
- c. The life span of corpus luteum of pregnancy is about 10 lunar months.
- d. Primary oocyte completes its first meiotic division at the time of fertilization.
- e. Nonciliated secretory cells present in the lining epithelium of uterine tube synthesise and store glycogen.

———21. Which of the following statements is **true** in the **female reproductive system**?

- a. The cells of theca interna secrete progesterone.
- b. Germinal (surface) epithelium covering the ovary gives rise to ova.
- c. Glassy membrane is the last structure to disappear in an atretic follicle.
- d. The alkaline environment of vagina prevents growth of microorganisms.
- e. The basic pH in the vaginal lumen is secretion of HCl by the tubuloalveolar glands of the vaginal walls.

———22. The **cornea**,

- a) constitutes the anterior half of the tunica fibrosa.
- b) is composed of 10 layers.
- c) is avascular.
- d) accounts for eye color.
- e) is attached to the ciliary body by zonules.

———23. In the **eye**, one of the following statements is **false**.

- a) The ciliary body generates the aqueous humor.
- b) The lens does not contain blood vessels and nerves
- c) The choroid is the vascular, pigmented layer of the eye.
- d) Lacrimal canaliculi are lined by a stratified squamous epithelium.

e) The cornea is covered by an incomplete layer of pigmented cells and fibroblasts on its anterior surface.

———24. In the **ear**, one of the following statements is **false**.

- a) the scala vestibule and tympani communicate with each other at the helicotrema.
- b) Otoliths are normally found in the saccule and utricle. tttt
- c) tectorial membrane is secreted by the stria vascularis of the ampullae of semicircular canals. ffff
- d) endolymphatic sac contains phagocytic cells like macrophages and neutrophils.
- e) the structure of the internal ear that contains special auditory receptors is the organ of Corti.

———25. With regard to the **ear**, one of the following statements is **true**.

- a). Hair cells of the ear are bipolar neurons of the spiral ganglion.
- b) The middle ear contains modiolus, cristae ampullaris and semicircular canals.
- c) Pinna consists of an irregular piece of elastic cartilage surrounded by a thick perichondrium.
- d) The lining epithelium of the outer aspect of the tympanic membrane is ciliated columnar interspersed with ceruminous glands.
- e. the bony labyrinth is filled with endolymph.

———49. Which of the following hormones is carried to its target(s) by the **hypophyseal portal system**?

- a) ADH
- b) ACTH
- c) FSH
- d) TRH
- e) TSH

———50. Which of the following hormones is **most** responsible for **increasing osteoclast number and activity?**

- a) ADH
- b) calcitonin
- c) estrogen
- d) parathyroid hormone
- e) testosterone

———51. Concerning the ***chromophobes of pars distalis of adenohypophysis***, which statement is **false**?

- a. may represent stem cells that have not yet differentiated into hormone-producing cells.
- b. maintain favorable fluid and electrolyte environment for the chromophil secretory cells.
- c. may be acidophils or basophils that are degranulated and thereby are depleted of hormone.
- d. do not stain intensely.
- e. may have minimal or no hormonal content.

———52. Which statement is **false** with regard to the ***epiphysis cerebri {pineal body}***?

- a. Melatonin may "delay" puberty through antigonadotropic effects.
- b. Its secretory activity is stimulated by darkness and inhibited by light.
- c. brain sand or corpora arenacea increase in size and number with increasing age.
- d. is surrounded by pia mater, which functions as its capsule. is covered by pia mater.
- e. None of the above.

———53. Which statement is **true** with regard to the ***endocrine system***?

- a. Myxedema is considered to be an autoimmune disease caused by hyperthyroidism.
- b. Calcitonin, secreted by the parathyroid gland, stimulates calcium resorption in the bones.
- c. The small and most numerous oxyphil cells of the parathyroid glands stain strongly basophilic.
- d. Both zona fasciculata and zona reticularis of the suprarenal gland depend on ACTH to sustain their function and survival.

e. The cells of the zona fasciculata of the suprarenal gland are called chromaffin cells because the granules of these cells can be stained strongly with potassium bichromate.

———54. The **neurohypophysis**,

- a) contains chromophils. b) includes the pars intermedia. c) secretes ACTH.
d) contains axons of hypothalamic neurons. e) hormone production is regulated by releasing factors.

———55. Cells specialized to secrete **steroids** typically contain:

- a. extensive smooth endoplasmic reticulum. b. numerous mitochondria with tubular cristae.
c. many lipid droplets. d. well-developed Golgi apparatus. e. all of the above.

———56. The **base of the hypothalamus** which connects with the **pituitary** is called the:

- a. anterior lobe. b. posterior lobe. c. neurohypophysis.
d. adenohypophysis. e. median eminence and infundibulum.

———57. The **pineal gland** is:

- a. derived evolutionarily from a medial photoreceptor organ.
b. located at the posterior end of the third ventricle.
c. innervated by sympathetic axons from the superior cervical ganglion.
d. composed of glial cells and cells (pinealocytes) similar to neurons. e. all of the above.

———58. Which tissue is most similar to a **sympathetic ganglion**?

- a. testicular interstitium b. pancreatic islets c. adrenal cortex d. adrenal medulla e. ovarian medulla

———59. **Epithelial cells** are arranged into cords around large capillaries or **sinusoids** in all of the following **EXCEPT** the:

- a. parathyroid. b. adenohypophysis. c. adrenal cortex. d. adrenal medulla. e. pancreatic islets.

———60. **Pancreatic alpha cells** secrete:

- a. insulin. b. glucagon. c. somatostatin. d. pancreatic polypeptide. e. aldosterone.

———61. **Calcitonin** is secreted by:

- a. parathyroid chief cells. b. thyroid follicular cells. c. thyroid parafollicular cells.
d. cells of adrenal cortex. e. cells of adrenal medulla.

———62. **Estrogen** and **progesterone** are secreted by cells of the:

- a. adrenal medulla. b. adrenal cortex. c. parathyroid. d. ovary. e. adenohypophysis.

———63. **Testis** has the following coverings **except**:

- a. tunica adventitia b. tunica vasculosa c. tunica albuginea d. tunica vaginalis e. None

———64. In the **testis**, which statement is **true**?

- a. Interstitial cells (of Leydig) produce testicular fluid.
b. The seminiferous tubules are lined with a complex stratified epithelium.
c. The secondary spermatocytes are the largest cells of the spermatogenetic lineage.
d. The tunica vaginalis is thickened on the posterior surface to form the mediastinum testis.
e. During spermiogenesis, the spermatids quickly undergo the second meiotic division to give spermatozoa.

———65. Which of the following is **NOT** considered part of the **male genital duct system**?

- a. Rete testis b. Tubuli recti c. Seminal vesicles d. Ductus deferens e. Ductus epididymidis

———66. Which statement is **false** with regard to the **prostate gland**?

- a. It is the largest accessory structure of the male reproductive system.
- b. The glands of the prostate lie in a fibromuscular stroma, also referred to as a fibroelastic stroma.
- c. Corpora amylacea {prostatic concretions} are the spherical structures seen in some prostatic alveoli.
- d. It is composed of compound tubuloalveolar glands.
- e. It produces testicular androgens.

———67. Which of the following statements is **false** in the **male reproductive system**?

- a. The secretion of seminal vesicle is particularly rich in fructose.
- b. Secondary spermatocytes are smaller than the primary spermatocytes.
- c. The epithelium lining the ductus epididymis is pseudostratified columnar.
- d. Tunica albuginea is the thickest in epididymis.
- e. The subcutaneous tissue in penis is devoid of fat.

———68. The function of **Sertoli cells** includes all of the following **EXCEPT**:

- a. phagocytosis of residual bodies from maturing spermatids.
- b. establishment through tight junctions of a blood/testis barrier.
- c. secretion of testosterone.
- d. metabolic and mechanical support of germ cells.
- e. secretion of tubular fluid.

———69. A network of tubules within the testis which interconnects **the seminiferous tubules with the efferent ductules** is the:

- a. epididymis.
- b. vas deferens.
- c. rete testis.
- d. tunica albuginea.
- e. spermatic cord.

———70. During **spermiogenesis**, the **spermatid** does all of the following **EXCEPT**:

- a. acquire an acrosome.
- b. acquire a flagellum.
- c. condense the nuclear chromatin.
- d. shed the euchromatin into the residual body.
- e. shed excess cytoplasm into the residual body.

———71. The **epithelial lining of the duct of the epididymis** is:

- a. simple squamous epithelium.
- b. ciliated columnar epithelium with goblet cells.
- c. transitional epithelium.
- d. pseudostratified epithelium with stereocilia.
- e. columnar epithelium with brush border and goblet cell

———72. **Functions for the epididymis** include all of the following **EXCEPT**:

- a. provide a site for sperm maturation.
- b. secrete various factors into the tubular lumen.
- c. provide a site for sperm storage.
- d. absorb excess fluid entering from the testicular tubules.
- e. secrete the bulk of ejaculate volume.

———73. The **seminal vesicle** consists of:

- a. serous secretory alveoli and branching ducts in a fibrous stroma.
- b. mucous secretory tubules.
- c. a coiled tubule lined by ciliated pseudostratified columnar epithelium.
- d. a hollow tube with an extensively folded mucosal lining, in a fibromuscular stroma.
- e. lymph nodules with crypts leading into the vas deferens.

———74. The **ejaculatory duct** connects the:

- a. testis with the epididymis.
- b. epididymis with the seminal vesicle.
- c. seminal vesicle with the prostatic urethra.
- d. prostatic urethra with the bladder.
- e. prostatic urethra with the penile urethra.

———75. **Semen** contains fluid derived from the testes and all of the following **EXCEPT**:

- a. epididymis and vas deferens. b. seminal vesicles. c. bulbourethral glands.
d. prostate. e. corpus spongiosum and corpora cavernosa.

———76. The **prostate**:

- a. contains glandular tissue in three layers: mucosal (periurethral), submucosal, and peripheral (main) glands.
b. contains secretory epithelium forming complex folds supported by lamina propria.
c. is lined by simple columnar or pseudostratified epithelium.
d. may contain small concretions (corpora amylacea) in glandular lumens. e. all of the above.

———77. The **penile urethra** passes through the:

- a. spermatic cord. b. corpora cavernosa. c. corpus spongiosum. d. prostate. e. seminal vesicle.

———78. The spaces within erectile tissue of **corpora cavernosa and corpus spongiosum** are:

- a. vascular, lined by endothelium. b. lymphatic, lined by endothelium. c. interstitial, with no proper lining.
d. peritoneal, lined by mesothelium. e. urinary, lined by transitional epithelium.

———79. Which organ functions as a site where **sperm cells mature after spermiogenesis**?

- a. testis b. epididymis c. vas deferens d. seminal vesicle e. prostate

———80. Which organ may **enlarge with age**, eventually causing **problems with urination**?

- a. testis b. epididymis c. vas deferens d. seminal vesicle e. prostate

———81. Which of the following is true of the **ampulla** of the **uterine tube** but **not** of its **isthmus**?

- a) Contains large mucosal folds that are extensively branched.
b) Contains smooth muscle in its walls.
c) Is covered by serosa derived from the peritoneum.
d) Contains ciliated epithelial cells.
e) Contains mucus-secreting peg cells.

———82. Which of the following statements is **true** about the **female reproductive system**?

- a. Mucosa of the uterus is commonly called perimetrium.
b. Fornix of vagina is lined by stratified squamous epithelium.
c. The life span of corpus luteum of pregnancy is about 10 lunar months.
d. Primary oocyte completes its first meiotic division at the time of fertilization.
e. Nonciliated secretory cells present in the lining epithelium of uterine tube synthesise and store glycogen.

———83. Which of the following statements is **true** in the **female reproductive system**?

- a. The cells of theca interna secrete progesterone.
b. Germinal (surface) epithelium covering the ovary gives rise to ova.
c. Glassy membrane is the last structure to disappear in an atretic follicle.
d. The alkaline environment of vagina prevents growth of microorganisms.
e. The basic pH in the vaginal lumen is due to the secretion of HCl by the tubuloalveolar glands of the vaginal walls.

———84. During **ovarian follicle maturation**, surrounding stromal cells differentiate to form the:

- a. granulosa. b. corona radiata. c. antrum. d. theca interna and theca externa. e. zona pellucida.

———85. **Ovarian follicles** which are degenerating after failing to mature are called:

- a. primordial follicles.
- b. primary / secondary follicles.
- c. tertiary / Graafian follicles.
- d. atretic follicles.
- e. corpora albicans.

———86. After ovulation, **granulosa cells** and cells of theca interna grow and differentiate to form the:

- a. corpus albicans.
- b. corpus luteum.
- c. zona pellucida.
- d. corona radiata.
- e. cumulus oophorus.

———87. The **corpus luteum** eventually degenerates, leaving a scar named the "**white body**" or:

- a. corpus albicans.
- b. atretic follicle.
- c. zona pellucida.
- d. corona radiata.
- e. cumulus oophorus.

———88. **Cilia** are found:

- a. at the surface of the germinal epithelium.
- b. on the surface of the ovum.
- c. on the free surface of corona radiata cells.
- d. lining the fallopian tube.
- e. lining the uterus.

———89. The **mucosal layer of the uterus** is called the:

- a. endometrium.
- b. myometrium.
- c. chorionic villi.
- d. placenta.
- e. cervix.

———90. The **proliferative phase of the endometrium** is characterized by all of the following **EXCEPT**:

- a. common mitotic figures in the endometrial stroma.
- b. common mitotic figures in the glandular epithelium.
- c. absence of differentiated secretory cells in the glandular epithelium.
- d. relatively smooth epithelial lining of the glandular tubules.

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- e. a stratified squamous surface.

———90. The fluid immediately beneath the cornea (i.e., between the cornea and the lens) is called:

- a. aqueous humor.
- B. vitreous humor.
- C. endolymph.
- D. perilymph.
- E. CSF.

———90. Aqueous humor is actively secreted by cells of the:

- a. cornea.
- B. stria vascularis.
- C. iris.
- D. ciliary processes.
- E. ependyma.

———90. The ciliary body of the eye is an anterior region of the choroid layer which consists largely of:

- a. collagen.
- b. smooth muscle.
- c. aqueous humor.
- d. vitreous humor.
- e. parasympathetic ganglion cells.

———90. An imbalance between production and drainage of aqueous humor can cause:

- a. Ménière's disease.
- B. astigmatism.
- C. presbyopia.
- D. glaucoma.
- E. hydrocephalus.

———90. As is evident from its embryological origin, the lens consists of which tissue type?

- a. epithelial tissue
- b. connective tissue
- c. nervous tissue

d. muscle tissue

_____90. Which layer of the retina contains nuclei of photoreceptor cells?

- a. pigmented epithelium b. receptor layer c. outer nuclear layer d. outer plexiform layer
e. inner nuclear layer

_____90. Which layer of the retina contains nuclei of bipolar cells?

- a. pigmented epithelium
b. receptor layer
c. outer nuclear layer
d. outer plexiform layer
e. inner nuclear layer
f. inner plexiform layer
g. ganglion cell layer
h. nerve fiber layer

_____90. Which layer of the retina contains synapses between photoreceptor cells and bipolar cells?

- a. pigmented epithelium
b. receptor layer
c. outer nuclear layer
d. outer plexiform layer
e. inner nuclear layer
f. inner plexiform layer
g. ganglion cell layer
h. nerve fiber layer

_____90. Which layer of the retina contains the blood vessels which serve the neural retina?

- a. pigmented epithelium
b. receptor layer
c. outer nuclear layer
d. outer plexiform layer
e. inner nuclear layer
f. inner plexiform layer
g. ganglion cell layer
h. nerve fiber layer

_____90. Which layer of the retina is responsible for absorbing stray light and prevents back-reflection?

- a. pigmented epithelium
b. receptor layer
c. outer nuclear layer
d. outer plexiform layer
e. inner nuclear layer
f. inner plexiform layer

- g. ganglion cell layer
- h. nerve fiber layer

_____90. Which layer of the retina recycles photoreceptor components?

- a. pigmented epithelium
- b. receptor layer
- c. outer nuclear layer
- d. outer plexiform layer
- e. inner nuclear layer
- f. inner plexiform layer
- g. ganglion cell layer
- h. nerve fiber layer

_____90. Which layer of the retina contains axons which travel through the optic nerve?

- a. pigmented epithelium
- b. receptor layer
- c. outer nuclear layer
- d. outer plexiform layer
- e. inner nuclear layer
- f. inner plexiform layer
- g. ganglion cell layer
- h. nerve fiber layer

_____90. Hair cells of the ear are:

- a. columnar mechanoreceptor cells.
- b. cuboidal cells of the stria vascularis.
- c. bipolar neurons of the spiral ganglion.
- d. bipolar neurons of the vestibular ganglion.
- e. ciliated cells of the eustachian tube.

_____90. Hair cells are found in ALL of the following locations EXCEPT:

- a. maculae of saccule.
- b. maculae of utricle.
- c. ampullae of semicircular canals.
- d. organ of Corti.
- e. stria vascularis.

_____90. The membranous labyrinth comprises ALL of the following EXCEPT:

- a. saccule.
- b. utricle.
- c. semicircular canals.
- d. scala media.
- e. scala tympani.

_____90. The **membranous labyrinth** is filled with:

- a. lymph.
- b. endolymph.
- c. perilymph.
- d. aqueous humor.
- e. CSF.

_____90. With regard to the histology of the **ear**, select the **incorrect statement**.

- a. Endolymph is actively secreted by cells of the stria vascularis.
- b. Endolymph can drain away from the membranous labyrinth through arachnoid granulations.
- c. Scala vestibuli contains perilymph.
- d. Otoliths are normally found in the saccule and utricle.
- e. A crista with a cupola is a normal feature in the ampulla of each semicircular canal.

_____90. The basilar and tectorial membranes are most closely associated with:

- a. secreting endolymph.
- b. reabsorbing endolymph.
- c. sensing linear acceleration of the head (i.e., gravity).
- d. sensing angular acceleration of the head.
- e. sensing vibratory stimuli (sound).

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_____90. In the organ of Corti, apical projections of hair cells (cilia and stereocilia) are in intimate contact with:

- a. Reissner's membrane.
- b. Decemet's membrane.
- c. the basilar membrane.
- d. the tectorial membrane.
- e. ciliary processes.

_____90. Hair cells in the organ of Corti synapse onto nerve fibers whose cell bodies are located in the:

- a. vestibular ganglion.
- b. spiral ganglion.
- c. cochlear nuclei.
- d. vestibular nuclei.
- e. thalamus.

_____90. Reissner's membrane separates the scala media from the:

- a. scala vestibuli.
- b. scala tympani.
- c. vestibule.
- d. eustachian tube.
- e. mastoid air cells.

_____90. The basilar membrane separates the scala media from the:

- a. scala vestibuli.
- b. scala tympani.
- c. vestibule.
- d. eustachian tube.
- e. mastoid air cells.

_____90. The round window separates the middle ear from the:

- a. scala vestibuli.
- b. scala tympani.
- c. scala media (cochlear duct).

- d. eustachian tube.
- e. mastoid air cells.

_____90. The middle ear communicates via the eustachian tube with the:

- a. posterior fossa.
- b. vestibule.
- c. nasopharynx.
- d. endolymphatic sac.
- e. external auditory meatus (ear canal).

HISTOLOGY

1. Which type of connective tissue makes the stroma of the visceral organs?

- a) White adipose tissue
- b) Brown adipose tissue
- c) Reticular connective tissue
- d) Elastic connective tissue
- e) Muroid connective tissue

2. Which connective tissue cell type is functionally related to the basophiles?

- a) Fibroblast
- b) Myofibroblast
- c) Histiocyte
- d) Plasma cell
- e) Mast cell

3. With regard to cartilage, which statement is **false**?

- a) Hyaline cartilage is characterized by an amorphous glassy matrix.
- b) Fibrocartilage has thick bundles of collagen fibres in the matrix.
- c) Fibrocartilage is present in regions subjected to high tension.
- d) Elastic cartilage is similar to hyaline cartilage except the abundant elastic fibres it contains.
- e) Fibrocartilage is used for endochondral bone ossification.

4. Which type of bone cells are involved in bone resorption?

- a) Osteoclast
- b) Osteocyte
- c) Osteoblast
- d) All of the above
- e) None of the above

5. Regions of cell bodies in the central nervous system make all of the following, **except**:

- a) Gray matter
- b) Column
- c) Layer
- d) Nuclei
- e) None of the above

6. Multipolar neurones appear under the light microscope:

- a) Spherical
- b) Angular
- c) Oval
- d) Spindle
- e) All of the above

7. Which of the following statements is true about the function of astrocytes?

- a) Act as supporting frame work
- b) Act as barrier for transmitters
- c) Store glycogen
- d) All of the above
- e) None of the above

8. Cells of epidermis that form the stratified squamous keratinized epithelium are:

- a) Keratinocytes
- b) Melanocytes
- c) Langerhan's cells
- d) Merckel's cells
- e) Chondrocyte

9. Sebaceous gland is a type of:

- a) Compound tubular gland.
- b) Compound alveolar gland.
- c) Simple straight tubular gland.
- d) Simple branched alveolar gland.
- e) Simple coiled tubular gland.

10. Which of the following statements is **not** correct regarding the veins?
- a) Are more in number than the arteries
 - b) Have prominent internal elastic lamina.
 - c) Have relatively well developed connective tissue and less developed muscles when compared with the arteries.
 - d) Have thinner wall and larger lumen when compared with the arteries.
 - e) Most veins have no clear boundary between tunica intima and media.
11. With regard to impulse generating and conducting system of the heart:
- a) Sinuatrial node is located on the lower aspect of the interatrial septal wall of the left atrium.
 - b) Cells of the sinuatrial node are larger in size when compared to those of the ordinary cardiac myocytes.
 - c) Cells of the sinuatrial node have more myofibrils than those of the ordinary cardiac myocytes.
 - d) Atrioventricular node is bigger in size than the sinuatrial node.
 - e) Purkinje cells, when seen under light microscopy following H & E staining, appear as large cells with clear zone surrounding centrally placed nucleus and denser peripheral cytoplasm.
12. Which of the following structures that does not make part of blood-thymus barrier:
- a) Endothelium
 - b) Type I epithelial reticular cells
 - c) Basal laminae
 - d) Smooth myocytes
 - e) None of the above
13. Which of the following statements is true regarding the lymphoid nodules (follicles)?
- a) Primarily made by accumulation of T lymphocytes
 - b) Have connective tissue capsule
 - c) Are found freely in connective tissue in some regions of the body or within the lymphoid organs
 - d) Primary lymphoid nodules show germinative region in the centre
 - e) All of the above
14. Which of the following statements is true about the lip?
- a) The inner aspect has sebaceous glands
 - b) The outer aspect has apocrine sweat glands
 - c) The transitional border has abundant salivary glands
 - d) The inner surface is lined by stratified columnar epithelium
 - e) The core contains skeletal muscles
15. Which of the following statements is true regarding the teeth?
- a) Odontoblastic processes branch and taper as they approach dentinoenamel junction.
 - b) Dentin is not present in the deciduous teeth.
 - c) Enamel has more collagen fibres than dentin.
 - d) Enamel is found overlying dentin in both crown and root portion of the teeth.
 - e) Enamel develops from mesoderm, while both dentin and cementum develop from overlying ectoderm.
16. Muscularis mucosae of esophagus:
- a) Is made by inner oblique, middle circular and outer longitudinal layers
 - b) Is made by inner circular and outer longitudinal layers
 - c) Is found only as longitudinal layer
 - d) Is entirely of skeletal muscle type
 - e) None of the above
17. Which of the following statements is true about the stomach?
- a) Gastric glands of cardiac region are compound branched tubuloalveolar in type
 - b) Gastric glands of pyloric region have longer glands than the pit with roughly 4:1 ratio
 - c) Has mucosa lined with stratified squamous nonkeratinized epithelium.
 - d) The most inner layer of muscularis externa is arranged as inner circular layer
 - e) Cells of the gastric glands that produce HCl are eosinophilic in staining

18. Which of the following statements is true about the small intestine?
- a) Has plicae circularis made by transverse folds of the circular muscle layer of muscularis externa
 - b) Has microvilli made by projections from absorptive cells
 - c) Has its most outer layer made by partly serosa and partly adventitia
 - d) Has its muscularis mucosae made by skeletal muscle
 - e) Has plicae circularis that increase the surface area more than microvilli or villi
19. Which of the following is correct about the histology of pancreas?
- a) Although is a mixed exocrine and endocrine gland, the majority of the secretory cells are exocrine cells.
 - b) The exocrine cells are arranged as simple tubular type.
 - c) The islets of Langerhans are made by larger and more eosinophilic cells than the exocrine cells.
 - d) The exocrine portions have more capillaries than the endocrine portion.
 - e) The B cells of islets of Langerhan's are distinguished under light microscope by being larger and more basophilic than the other cells of the islets.
20. Which of the following type of cells is **not** present in the respiratory epithelium of the nasal cavity?
- a) Ciliated columnar cells
 - b) Goblet cells
 - c) Basal cells
 - d) Stem cells
 - e) None of the above
21. In which of the following regions of the respiratory tract do we find stratified squamous nonkeratinized epithelium?
- a) Larynx
 - b) Principal bronchi
 - c) Tertiary bronchi
 - d) Trachea
 - e) Bronchioles.
22. Which type of neurones of the cerebral cortex are predominantly found in the primary sensory cortex?
- a) Stellate cells
 - b) Horizontal cells of Cajal
 - c) Fusiform cells
 - d) Cells of Martinotti
 - e) Pyramidal cells
23. The correct order of the three layers of cerebellar cortex, outer to inner is:
- a) Molecular layer, Purkinje cell layer and granular layer
 - b) Molecular layer, granular layer and Purkinje cell layer
 - c) Purkinje cell layer, molecular layer, and granular layer
 - d) Granular layer, molecular layer, Purkinje cell layer
 - e) Granular layer, Purkinje cell layer and molecular layer
24. Which of the following statements is **not true** regarding the meninges?
- a) Cranial dura mater has an outer layer which is connected by Sharpey's fibres to the calvaria
 - b) Dura mater of the spinal cord is made up of only one layer of dense fibrous membrane
 - c) Arachnoid mater is delicate and transparent impermeable membrane
 - d) Arachnoid mater has numerous trabeculae which are connected to the dura mater traversing the subdural space
 - e) Arachnoid granulations are sites of formation of cerebrospinal fluid into the ventricle
25. Which of the following is involved in the formation of blood brain barrier?
- a) Astrocytes
 - b) Ependymal cells
 - c) Oligodendrocytes
 - d) Microglia
 - e) Schwann cells

II. EMBRYOLOGY

Each of the numbered items or incomplete statements is followed by answers or by completions of the statement. Select the one lettered answer or completion that is best in each case. (Total : 60 points)

- _____ 1. The **anterior 2/3rd of the tongue** develops at which pharyngeal arch level?
a) 1st b) 2nd c) 3rd d) 4th e) 6th
- _____ 2. Which of the following **salivary glands** develop from ectoderm?
a) Parotid b) Submandibular c) Sublingual d) Lingual e) All of the above
- _____ 3. All of the following **appendages of the skin** develop from ectoderm, **except**:
a) Hairs. b) Sweat glands. c) Sebaceous glands. d) Arrector pili muscle. e) Nails.
- _____ 4. Concerning the **development of the GIT and associated structures**, which statement is **true**?
a) Liver develops by fusion of ventral and dorsal hepatic buds.
b) Stomach rotates 270° counterclockwise, similar to the midgut.
c) The loop of the midgut has the superior mesenteric artery in its axis.
d) The hepatocytes of liver are of mesodermal in origin, while the stroma is of endodermal origin.
e) The exocrine portion of pancreas develops from endoderm of the gut, while the endocrine portion. develops from the local mesoderm.
- _____ 5. With regard to the **congenital anomalies of the kidneys and ureters** which statement is **false**?
a) Accessory renal arteries are end arteries.
b) Bilateral renal agenesis is associated with oligohydramnios.
c) Incontinence is a common complaint resulting from an ectopic ureteric orifice in both sexes.
d) Unilateral renal agenesis is more common in males than females. e) None of the above.
- _____ 6. Which of the following is not the **adult derivative of the ureteric bud**?

- a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

_____ **7. Urachal cyst** is a remnant of:

- a) urogenital sinus b) urogenital rigde c) cloaca d) allantois e) mesonephric duct

_____ **8. The podocytes of Bowman's capsule** are derived from:

- a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

_____ **9. With regard to the development of the endocrine system,** which of the following statements is **false**?

- a) the oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.
b) C cells differentiate from neural crest cells that migrate from the pharyngeal arches into the 4th pair of pharyngeal pouches.
c) the thyroid gland is the first endocrine gland to develop in the embryo.
d) the pituitary gland is ectodermal in origin.
e) the pineal gland develops as a median diverticulum of the caudal part of the roof of the diencephalon.

_____ **10. A structure found within the adult female pelvis formed from the gubernaculum** is

- a) broad ligament b) superior ligament of the ovary c) medial umbilical ligament
d) round ligament of the uterus e) median umbilical ligament

_____ **11. During the development of the kidney systems,** which statement is **true**?

- a) The transitional epithelium lining the urethra is derived from neuroectoderm.
b) The "ascent" of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
c) The pronephros are derived from intermediate mesoderm from upper thoracic to upper lumbar segments.
d) Urine production begins shortly after birth. e) At birth, the kidneys have a lobulated appearance.

_____ **12. The labia minora** arise embryologically from:

- a) phallus b) labioscrotal swellings c) sinovaginal bulbs d) urogenital folds e) paramesonephric duct

_____ **13. The uterine tubes** of the adult female are derived embryologically from

- a) mesonephric duct b) mesonephric tubules c) paramesonephric duct
d) paramesonephric tubules e) uterovaginal primordium

_____ **14. The most common cause of testicular feminization [Androgen Insensitivity] syndrome** is

- a) a 46, XO genotype b) a 47, XXY genotype c) lack of androgen receptors
d) congenital adrenal hyperplasia e) inadequate production of testosterone and müllerian-inhibiting factor.

_____ **15. In the male, failure of urethral folds to fuse completely** results in

- a) hypospadias b) epispadias c) cryptorchidism d) hydrocele e) congenital inguinal hernia

_____ **16. Intermediate mesoderm** gives rise to the

- a) neural tube b) heart c) kidneys and gonads d) somites e) notochord

_____ **17. Concerning the development of the Urinary System,** which statement is **false**?

- a) The kidneys attain their definitive adult position postnatally.
b) Urinary system begins to develop before the genital system.
c) The metanephric diverticulum is the primordium of the ureter.

- d) Nephrons are derived from the metanephric mass of intermediate mesoderm.
- e) Differentiation of the nephrons depends upon induction by the collecting tubules.

_____ 18. Which of the following is not the adult derivative of the **ureteric bud**?

- a) renal pelvis
- b) major calyx
- c) minor calyx
- d) collecting duct
- e) connecting tubule

_____ 19. Concerning the **development of the urinary bladder and urethra**, which statement is **false**?

- a) The female urethra is derived from endoderm of the urogenital sinus.
- b) In adults, the urachus is represented by the median umbilical ligament.
- c) The urinary bladder becomes a pelvic organ during the 9th week of development.
- d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
- e) Epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

_____ 20. Up to the 5th week of development, the embryonic **skin** is only a single layer of:

- a) Ectoderm
- b) Endoderm
- c) Paraxial mesoderm
- d) Intermediate mesoderm
- e) Lateral plate mesoderm

_____ 21. During the development of the **female reproductive system**, which statement is **false**?

- a) The urogenital folds remain unfused and form the labia minora.
- b) The cranial portion of the paramesonephric ducts form the uterine tubes.
- c) Didelphys (double) uterus anomalies occur when there is a complete lack of fusion of paramesonephric ducts.
- d) The phallus forms the mons pubis.
- e) The labioscrotal swellings form the labia majora.

_____ 22. Concerning the **development of the genital system**, one statement is **false**?

- a) Bilateral cryptorchidism may result in sterility.
- b) Hypospadias is constantly associated with exstrophy of urinary bladder.
- c) In testicular feminization syndrome, uterus and uterine tubes are generally absent.
- d) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or labia majora
- e) In epispadias usually the penis is underdeveloped and is curved ventrally, a condition known as chordee.

_____ 23. With regard to the **development of gonads**, which statement is **false**?

- a) The surface epithelium of the female gonad develop into oogonia.
- b) Sex of an embryo is determined genetically at the time of fertilization.
- c) Germ cells do not appear in the genital ridges until the 6th week of development.
- d) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.
- e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

_____ 24. With regard to the **vestigial structures derived from embryonic genital ducts**, which statement is **false**?

- a) The prostatic utricle is homologous to the vagina.
- b) The cranial end of mesonephric duct in males may persist as paradidymis.
- c) The epoophoron in the females correspond to the efferent ductules and duct of epididymis
- d) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
- e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

_____ 25. Concerning the development of the **Integumentary system**, which statement is **true**?

- a) castoff cells from the epidermis mix with secretions of sebaceous glands to form the vernix caseosa.
- b) toenails and fingernails begin to develop at the tips of the digits postnatally.
- c) teeth develop from ectoderm, mesoderm and neural crest cells.
- d) eccrine sweat glands develop from mesenchyme, which is derived from the mesoderm underlying the surface ectoderm.
- e) the epidermis is derived from surface ectoderm.

_____ **26.** During the **development of the digestive system**, which statement is **true?**

- a) omphalocele is a persistence of the herniation of abdominal contents into the proximal part of the umbilical cord.
- b) the superior two-thirds of the adult anal canal are derived from proctodeum.
- c) the midgut may give rise to the descending colon and sigmoid colon.
- d) the dorsal pancreatic bud forms the uncinat process and part of the head of the pancreas.
- e) the anal membrane normally breaks down by the end of the second trimester.

_____ **27.** What germ layer forms the melanocytes?

- a) Endoderm
- b) Neural crest
- c) Paraxial mesoderm
- d) Intermediate mesoderm
- e) Lateral plate mesoderm

_____ **28.** What germ layer is the **dermis** derived from?

- a) Paraxial mesoderm
- b) Intermediate mesoderm
- c) Lateral plate mesoderm
- d) Neural crest
- e) Endoderm

_____ **29.** At birth the infant is covered by a whitish paste (vernix caseosa), which protects the skin against the macerating action of amniotic fluid. Which of the following is the **vernix caseosa** derived from?

- a) Arrector pili muscles
- b) Dermal root sheaths
- c) Hair papilla
- d) Sebaceous glands
- e) Hair blood vessels

_____ **30.** During a routine examination of a female patient, **accessory nipples** are found in the axilla and on the mid thorax along the embryological mammary ridge. What is the terminal extension of this ridge?

- a) Mid thorax
- b) Diaphragm
- c) Umbilicus
- d) Pubic region
- e) Lower thigh to knee

_____ **31.** From what layer of the epidermis do **hair follicles** begin to proliferate?

- a) Stratum corneum
- b) Stratum lucidum
- c) Stratum granulosum
- d) Stratum spinosum
- e) Stratum germinativum

_____ **32.** This disorder results from **excessive keratinization of the skin**, whereby the skin appears as very dry and scaly. In extreme cases (harlequin fetus), the skin is abnormally thick and cracked.

- a) Albinism
- b) Amelogenesis imperfect
- c) Leprosy
- d) Psoriasis
- e) Ichthyosis

_____ **33.** A patient presents with **red blotches on their face and neck**. They say they've had these birthmarks as long as they can remember and they were told it has something to do with capillaries. Which of the following malformation does this patient likely have?

- a) Albinism
- b) Angioma
- c) Ichthyosis
- d) Collodion
- e) Thalidomide defect

_____ **34.** The **midgut** begins with _____ and ends with _____.

- a) 1/2 jejunum; 2/3 transverse colon
- b) 1/2 ileum; 2/3 sigmoid colon
- c) 1/2 jejunum; 2/3 rectum
- d) 1/2 duodenum; 2/3 transverse colon
- e) 1/2 duodenum; 1/3 sigmoid colon

- _____ **35.** The **middle of the duodenum** would receive blood from:
a) Superior mesenteric artery only b) Inferior mesenteric artery only c) Celiac artery only
d) Celiac and inferior mesenteric artery e) Celiac and superior mesenteric artery
- _____ **36.** What two structures only form on the **ventral side of the bile duct**?
a) Liver; Pancreas b) Liver; Gallbladder c) Pancreas; Gallbladder
d) Heart; Liver e) Heart; Pancreas
- _____ **37.** During week six, the **midgut herniates** through the umbilical ring, rotating _____ degrees counter clockwise. Near week ten, the midgut returns, rotating _____ degrees for an overall rotation of _____ degrees.
a) 180; 180; 0 b) 180; 180; 360 c) 90; 180; 90 d) 90; 180; 270 e) 270; 90; 180
- _____ **38.** A three-week old male infant comes into the Emergency Room for **weight loss and acute dehydration**. The infant has been **projectile vomiting** and the condition is worsening. Which of the following is most likely malformation?
a) Gastric atresia b) Pyloric stenosis c) Annular pancreas
d) Malrotation of the gut e) Intestinal duplication
- _____ **39.** **Meckel's diverticulum** is a congenital connection from the umbilicus via a vitelline ligament to _____
a) Duodenum b) Jejunum c) Ileum d) Transverse colon e) Sigmoid colon
- _____ **40.** During the **pronephros period** (week 3), the first kidney develops in the _____ region. By week 4 this kidney degenerates.
a) Cervical b) Thoracic c) Lumbar d) Sacral e) Cloaca
- _____ **41.** The **ureteric bud** gives rise to the _____ and the **metanephric mesoderm** gives rise to the _____.
a) Nephrons; Collecting system b) Collecting system; Nephrons c) Bowman's capsule; Ureter
d) Loop of Henle; Renal pelvis e) Distal/proximal convoluted tubules; Major/minor calyces
- _____ **42.** The **urorectal septum** divides the _____ from the _____.
a) Anorectal canal; Urogenital sinus b) Urogenital sinus; Mesonephric duct c) Urachus; Allantois
d) Urinary bladder; Allantois e) Mesonephric duct; Ureteric bud
- _____ **43.** The **urachus** forms the:
a) Allantois b) Median umbilical ligament c) Medial umbilical ligament
d) Lateral umbilical ligament e) Round (umbilical) ligament
- _____ **44.** The **bladder** becomes continuous with the lower end of the mesonephric (Wolffian) duct, which will form the _____ of the bladder.
a) Urethra b) Prostate c) Trigone d) Posterior wall e) Anterior wall
- _____ **45.** At birth, a child has **large palpable flank masses and abnormal extremities**. The delivery physician is very concerned as she knows this **autosomally-dominant disorder** can lead to renal failure and the child will likely need a kidney transplant to survive. Which of the following disorders does the child likely have?
a) Polycystic kidney b) Renal agenesis c) Pelvic kidney d) Bifid ureter e) Bifid kidney
- _____ **46.** This disorder, associated with **oligohydramnios**, occurs when the ureteric bud fails to reach the metanephric mesoderm.

- a) Polycystic kidney b) Renal agenesis c) Pelvic kidney
d) Horseshoe kidney e) Bifid ureter

_____ 47. What artery prevents the **horseshoe kidney** (metanephric mesodermal fusion) from ascending high into the abdomen?

- a) Celiac trunk b) Superior mesenteric artery c) Inferior mesenteric artery
d) Left renal artery e) Right renal artery

_____48. A newborn baby presents with fluid draining from the **umbilicus onto the skin**. Testing of the fluid identifies it as urine. What is the most likely diagnosis?

- a) Urachal cyst b) Urachal fistula c) Exstrophy of the bladder d) Bifid ureter e) Horseshoe kidney

_____ 49. A patient is born with karotype **46, XY** but has no problem with production of testosterone or müllerian inhibiting hormone. Their **vagina ends in a blind pouch** and **testes are lodged in the inguinal canal**. Which of the following is the best description of this patient:

- a) Male pseudo-hermaphrodite b) Female pseudo-hermaphrodite c) Androgen insensitivity syndrome
d) Turner's syndrome e) Klinefelter syndrome

_____ 50. During development of the **ovaries**, the **primitive sex (medullary) cords** _____.

- a) Proliferate b) Move laterally c) Move medially d) Differentiate e) Degenerate

_____51. The **gubernaculum** is attached to the uterus and as the ovaries descend it becomes the _____
_____of the uterus.

- a) Canal of Nuck b) Ovarian ligament c) Round ligament
d) Ovarian and round ligaments e) Gubernacular ligament

_____ **52. Development of the testis** is induced by the sry-gene, which encodes testis-determiningfactor (TDF). TDF causes all of the following **EXCEPT**:

- a) The sex cords continue to proliferate to form the testis.
- b) Leydig cells are secreting testosterone thus have very little SER.
- c) Towards the hilum, the cords grow into a network that will give rise to the rete testes.
- d) During further development, a thick layer (tunica albuginea) separates the cords from the surface epithelium.
- e) Sex cords become seminiferous cords that form seminiferous tubules (Sertoli cells), straight tubules, and rete testes

_____ **53.** Which of the following inhibits the growth of the **paramesonephric duct (males)**?

- a) Testosterone b) Leydig Cells c) Dihydrotestosterone d) Estrogens e) Müllerian substances (MIFs)

 54. As the **testes descend**, they follow the:

- a) Processus vaginalis b) Parietal layer of tunica vaginalis c) Visceral layer of tunica vaginalis
d) Caudal genical ligament e) Cryptorchidism

55. The epididymis, ductus deferens, ejaculatory duct, and seminal vesicle all arise from:

- a) The Müllerian ducts b) The paramesonephric ducts c) The mesonephric ducts
d) Leydig hormones e) Wolffian hormones

56. The uterine tubes, uterus, cervix, and superior part (1/3) of the vagina all arise from:

- a) The Wolffian ducts b) The paramesonephric ducts c) The mesonephric ducts

d) Leydig hormones

e) Müllerian hormones

_____ **57. The epoophoron and paroophoron, which are important as cysts, arise from:**

a) The Mullarian ducts

b) The paramesonephric ducts

c) The mesonephric ducts

d) Leydig hormones

e) Wolffian hormones

_____ **58. The lower portion of the vagina is derived from:**

a) Sinovaginal bulb

b) Uteral lumen

c) Hymen

d) Urogenital sinus

e) Vaginal plate

_____ **59. What structure in the indifferent stage (weeks 3-6) becomes the clitoris?**

a) Urogenital groove

b) Genital swelling

c) Urethral fold

d) Perineum

e) Genital tubercle

_____ **60. Genetic 46,XY would be:**

a) Turner's syndrome

b) Klinefelter syndrome

c) True hermaphrodite

d) Male pseudo-hermaphrodite

e) Female pseudo-hermaphrodite

_____ **1. Most birth defects occur during which of the following periods/stages of the intrauterine life?**

a) First 2 weeks

b) Embryonic period

c) Fetal period

d) Last trimester

e) Zygote

_____ **2. Which of the following statements is not true regarding the human placenta?**

a) Maternal portion is formed by decidua parietalis.

b) Fetal portion is formed by chorion frondosum.

c) Cotyledons are seen from the maternal aspect.

d) Fetal portion is smooth.

e) At full term covers the whole surface area of the endometrium.

_____ **3. The central tendon of the diaphragm develops from:**

a) Dorsal mesentery of the esophagus

b) Pleuroperitoneal membranes

c) Pleuropericardial membranes.

d) Lateral body wall

e) Septum transversum.

_____ **4. Which of the following is not a component of vernix caseosa?**

a) Secretions from sebaceous gland

b) Secretions of apocrine sweat glands

c) Fallen epidermal cells

d) Fallen hairs

e) None of the above

_____ **5. Structure that is not developed in the third branchial arch include:**

a) Lesser cornu of hyoid bone

b) Body of the hyoid bone

c) Glossopharyngeal nerve

d) Common carotid artery

e) Stylopharyngeus muscle

_____ **6. Which of the following structures develop from endoderm of the second pharyngeal pouch?**

a) Thymus gland

b) Palatine tonsil

c) Thyroid gland

d) Superior parathyroid gland

e) Parafollicular cells of thyroid gland

_____ **7. The anterior 2/3rd of the tongue develops at which pharyngeal arch level?**

a) 1st

b) 2nd

c) 3rd

d) 4th

e) 6th

_____ **8. Which of the following salivary glands develop from ectoderm?**

a) Parotid

b) Submandibular

c) Sublingual

d) Lingual

e) All of the above

_____ **9. Secretion of surfactant**, which is an important substance for survival of premature infants appears during which of the following periods of lung development?

- a) Canalicular. b) Bronchiolar. c) Terminal sac period d) Pseudoglandular e) Alveolar

_____ **10. Regarding the development of the gut**, which of the following statements is **correct**?

- a) The lumen of the intestine is temporarily blocked by epithelial proliferation.
b) The site of junction of foregut and midgut derivatives is the middle of jejunum.
c) Is formed as the result of absorption of part of the amniotic cavity because of the embryonic foldings.
d) The midgut is temporarily connected to the yolk sac by the allantois.
e) All of the above.

_____ **11. All of the following appendages of the skin** develop from ectoderm, except:

- a) Hairs. b) Sweat glands. c) Sebaceous glands. d) Arrector pili muscle. e) Nails.

_____ **12. With regard to fertilization**, which statement is **false**?

- a) restores the normal diploid number of chromosomes in the zygote.
b) results in variation of human species through mingling of maternal and paternal chromosomes.
c) causes metabolic activation of the ovum and initiates cleavage of the zygote.
d) determines chromosomal sex of the embryo. e) None of the above.

_____ **13. During the first week of development**,

- a) the embryo is a solid ball of blastomeres and is called gastrula.
b) mesoderm is formed which will give rise later to the placental membranes.
c) the blastocyst is superficially implanted in the uterine wall (endometrium).
d) developmental disturbances may give rise to major congenital anomalies.
e) the amniotic cavity is developed within the yolk sac as a space between the cytotrophoblast and the embryoblast.

_____ **14. During the third week of development**, which statement is **false**?

- a) somites, which will give rise to the vertebrae, ribs and axial musculature, are formed.
b) notochord forms the primitive axis of the embryo around which the axial skeleton will form.
c) the primitive streak appears as a thickening of the embryonic epiblast at the caudal end of the embryonic disc.
d) mesoderm exists between the ectoderm and endoderm everywhere. e) neural tube is formed.

_____ **15. During the fetal period**,

- a) the spleen is the major site of erythropoiesis during the end of the 9th week.
b) ossification of the fetal skeleton is active during the 13th -16th week.
c) the bodies of 20- week fetuses are usually completely covered with lanugo hairs.
d) by 18 weeks the uterus is formed and canalization of the vagina has begun.
e) the amount of white fat is about 16% of body weight at 29 weeks.

_____ **16. With regard to the placenta**,

- a) the maternal part of the placenta develops from the chorionic sac.
b) the fetal part of the placenta is attached to the maternal part of the placenta by the cytotrophoblastic shell.
c) amniochorionic membrane ruptures during labor.
d) as pregnancy advances, the placental membrane becomes progressively thinner.
e) the fetal surface of a freshly delivered placenta has the characteristic cobblestone appearance produced by cotyledons.

_____ **17. Concerning the fetal membranes (developmental adnexa),**

- a) the yolk sac helps control the embryo's body temperature.
- b) primordial germ cells appear in the endodermal lining of the wall of the allantois.
- c) during the 4th week the endoderm of the yolk sac is incorporated into the embryo as the primordial gut.
- d) as the bladder enlarges, the umbilical cord involutes to form urachus.
- e) blood formation occurs in the wall of allantois during the third to fifth weeks.

_____ **18. Concerning multiple pregnancies,**

- a) twins with one amnion, one chorion and one placenta are always monozygotic.
- b) the rate of dizygotic twinning increases with maternal age.
- c) the frequency of monozygotic twinning shows marked racial differences.
- d) dizygotic twins always have two amnions and two chorions.
- e) monozygotic twins are of the same sex.

_____ **19. The neuroepithelial cells of the neural tube gives rise to all of the following cells of the brain except:**

- a) Microglia.
- b) Astrocytes.
- c) Oligodendrocytes.
- d) Purkinje cells of cerebellar cortex.
- e) Ependymal cells.

_____ **20. With regard to birth defects,**

- a) the phenotype of Turner syndrome is female.
- b) thalidomide induces limb defects.
- c) trisomy of the autosomes occurs with increasing frequency as maternal age increases.
- d) deafness has been reported in infants of mothers who have been treated with penicillin during the 2nd trimester.
- e) phenobarbital is considered to be a safe, antiepileptic drug for use during pregnancy.

_____ **21. Concerning the development of the GIT and associated structures:**

- a) The loop of the midgut has the superior mesenteric artery in its axis.
- b) The exocrine portion of pancreas develops from endoderm of the gut, while the endocrine portion develops from the local mesoderm.
- c) Liver develops by fusion of ventral and dorsal hepatic buds.
- d) The hepatocytes of liver are of mesodermal in origin, while the stroma is of endodermal origin.
- e) Stomach rotates 270° counterclockwise, similar to the midgut.

_____ **22. With regard to the development of the musculoskeletal system,**

- a) the muscular system develops from neuroectoderm.
- b) the upper limbs rotate laterally through 90° on their longitudinal axes.
- c) premature closure of the frontal (metopic) suture results in scaphocephaly.
- d) sclerotomes of the somites form the viscerocranium.
- e) xiphoid process develops during the 6th week.

_____ **23. Cerebellum develops from cells of:**

- a) Basal plate of myelencephalon
- b) Alar plate of myelencephalon
- c) Basal plate of metencephalon
- d) Alar plate of myelencephalon
- e) Basal plate of mesencephalon

_____ **24. Primary brain vesicle that remains relatively unchanged into the secondary brain vesicle is:**

- a) Rhombencephalon
- b) Prosencephalon
- c) Telencephalon
- d) Mesencephalon
- e) Diencephalons

- [illegible]

- a) umbilical chord b) heart c) mouth d) anus e) nose

- a) neurulation b) gastrulation c) craniocaudal folding d) lateral folding e) angiogenesis

- a) both ectoderm and mesoderm b) neural crest c) endoderm
d) mesoderm e) surface ectoderm

- a) is of mesodermal origin b) develops from the otic vesicle c) develops from pharyngeal arch 1
d) develops from pharyngeal pouch 2 e) develops from pharyngeal pouch 1

- a) Accessory renal arteries are end arteries.
b) Bilateral renal agenesis is associated with oligohydramnios.
c) Incontinence is the common complaint resulting from an ectopic ureteric orifice in both sexes.
d) Unilateral renal agenesis is more common in males than females. e) None of the above.

- a) organ of Corti b) endolymphatic duct c) superior semicircular canal
d) crus commune nonampullare e) lateral semicircular canal

- a) cochlear duct b) ductus reuniens c) perilymphatic duct
d) vestibular aqueduct e) utriculosaccular duct

- a) retina
b) dilator and sphincter pupillae muscles of iris
c) corneal endothelium
d) vitreous body
e) lens

- a) renal pelvis b) major calyx c) minor calyx d) collecting duct e) connecting tubule

- a) urogenital sinus b) urogenital rigde c) cloaca d) allantois e) mesonephric duct

- a) ectoderm b) mesoderm c) endoderm d) endoderm and mesoderm e) neural crest cells

36. A structure found within the adult **female pelvis formed from the gubernaculum** is

- a) broad ligament b) superior ligament of the ovary c) medial umbilical ligament
d) round ligament of the uterus e) median umbilical ligament

_____ **37.** After the ***sinovaginal bulbs*** have proliferated and fused, they form a solid core of endodermal cells called

- a) vestibule of the vagina b) uterovaginal primordium c) urogenital sinus
d) vaginal plate e) clitoris

_____ **38.** The ***labia minora*** arise embryologically from

- a) phallus b) labioscrotal swellings c) sinovaginal bulbs
d) urogenital folds e) paramesonephric duct

_____ **39.** The ***uterine tubes*** of the adult female are derived embryologically from

- a) mesonephric duct b) mesonephric tubules c) paramesonephric duct
d) paramesonephric tubules e) uterovaginal primordium

_____ **40.** The most common cause of ***testicular feminization syndrome*** is

- a) a 46, XO genotype b) a 47, XXY genotype
c) lack of androgen receptors d) congenital adrenal hyperplasia
e) inadequate production of testosterone and müllerian-inhibiting factor (MIF)

_____ **41.** In the male, ***failure of urethral folds to fuse completely*** results in

- a) hypospadias b) epispadias c) cryptorchidism d) hydrocele e) congenital inguinal hernia

_____ **42.** The ***Y chromosome*** carries a gene on its short arm that codes for

- a) testosterone b) MIF c) testes-determining factor (TDF) d) progesterone e) estrogen

_____ **48.** Which of the following best describes the ***placental components of dizygotic twins***?

- a) one placenta, two amniotic sacs, one chorion b) one placenta, two amniotic sacs, two chorions
c) two placentas, two amniotic sacs, one chorion d) two placentas, two amniotic sacs, two chorions
e) one placenta, two amniotic sacs, two chorions

_____ **43.** The ***notochord*** induces the overlying ectoderm to develop into the

- a) skin b) eye c) primitive streak d) primitive node e) neural plate

_____ **44.** A ***sacrooccygeal teratoma*** is a tumor that arises from remnants of the

- a) neural plate b) cloacal membrane c) posterior neuropore d) primitive streak e) notochord

_____ **45.** ***Intermediate medoderm*** gives rise to the

- a) neural tube b) heart c) kidneys and gonads d) somites e) notochord

_____ **46.** *The proximal part of the right subclavian artery is derived from aortic arch*

- a) 1 b) 2 c) 3 d) 4 e) 6*

_____ **47.** *The portal vein is derived from*

- a) vitelline veins b) umbilical veins c) anterior cardinal veins
e) posterior cardinal veins e) subcardinal veins*

_____ **49.** The ***stapedius muscle*** that moves the stapes ossicle is innervated by

- a) CN V3 b) CN XII c) CN III d) CN VII e) cervical nerves C2 and C 3

_____ **50. Pharyngeal groove 1** gives rise to

- a) internal auditory meatus
- b) external auditory meatus
- c) eustachian tube
- d) cervical sinus
- e) primary tympanic cavity

_____ **51. Failure of choroid fissure to close** results in

- a) congenital detached retina
- b) congenital aniridia
- c) congenital aphakia
- d) coloboma iridis
- e) microphthalmos

_____ **52. Aqueous humor** is produced by the

- a) choroid plexus
- b) trabecular meshwork
- c) ciliary processes
- d) vitreous body
- e) lens vesicle

_____ **53. Concerning the development of the urogenital system**, which of the following statements is true?

- a) The labia minora arise embryologically from paramesonephric ducts.
- b) The uterine tubes of the adult female are derived embryologically from uterovaginal primordium.
- c) The most common cause of testicular feminization syndrome is the presence of abundant androgen receptors.
- d) Round ligament of the uterus is formed from the gubernaculum.
- e) The X chromosome carries a gene on its short arm that codes for TDF.

_____ **54. With regard to the development of the endocrine system**, which of the following statements is false?

- a) the oxyphil cells of parathyroid gland differentiate during the embryonic period and become functionally active in regulating fetal calcium metabolism.
- b) C cells differentiate from neural crest cells that migrate from the pharyngeal arches into the 4th pair of pharyngeal pouches.
- c) the thyroid gland is the first endocrine gland to develop in the embryo.
- d) the pituitary gland is ectodermal in origin.
- e) the pineal gland develops as a median diverticulum of the caudal part of the roof of the diencephalon.

_____ **55. The transitional epithelium lining the ureter** is derived from

- a) ectoderm
- b) mesoderm
- c) endoderm
- d) endoderm and mesoderm
- e) neural crest cells

_____ **56. Which of the following is not the adult derivative of the ureteric bud?**

- a) renal pelvis
- b) major calyx
- c) minor calyx
- d) collecting duct
- e) connecting tubule

_____ **57. Which of the following statements about the optic nerve is false? The optic nerve**

- a. is a tract of diencephalon.
- b. is myelinated by oligodendrocyte.
- c. is invested by meninges.
- d. is not capable of regeneration after transaction.
- e. is developed from the first pharyngeal pouch.

_____ **58. Concerning the development of the Urinary System**, which statement is false?

- a) Urinary system begins to develop before the genital system

- b) The metanephric diverticulum is the primordium of the ureter.
- c) Nephrons are derived from the metanephric mass of intermediate mesoderm.
- d) Differentiation of the nephrons depends upon induction by the collecting tubules.
- e) The kidneys attain their definitive adult position toward the end of the fetal period.

_____ **60.** Concerning the development of the urinary bladder and urethra, which statement is **false**?

- a) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
- b) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.
- c) In adults, the urachus is represented by the median umbilical ligament.
- d) The urinary bladder becomes a pelvic organ during the 9th week of development.
- e) The female urethra is derived from endoderm of the urogenital sinus.

_____ **63.** During the development of the kidney systems, which statement is **true**?

- a) The transitional epithelium lining the urethra is derived from neuroectoderm.
- b) The "ascent" of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.
- c) The pronephros are derived from intermediate mesoderm from upper thoracic to upper lumbar segments.
- d) Urine production begins shortly after birth.
- e) At birth, the kidneys have a lobulated appearance.

_____ **64.** During the development of the **female reproductive system**, which statement is **false**?

- a) The urogenital folds remain unfused and form the labia minora.
- b) The cranial portion of the paramesonephric ducts form the uterine tubes.
- c) Didelphys (double) uterus anomalies occur when there is a complete lack of fusion of paramesonephric ducts.
- d) The phallus forms the mons pubis.
- e) The labioscrotal swellings form the labia majora.

_____ **65.** With regard to the development of the male reproductive system, which statement is **false**?

- a) The genotype of an embryo is established at fertilization.
- b) By week 12, male characteristics of the external genitalia are recognizable.
- c) Sertoli cells are derived from mesoderm.
- d) Seminiferous tubules are formed by week 20.

e) In epispadias usually the penis is underdeveloped and curved ventrally a condition known as chordee.

_____ **66.** Concerning the **development of the genital system**, one statement is **false?**

- a) Hypospadias is constantly associated with exstrophy of urinary bladder.
- b) Congenital inguinal hernia occurs when a loop of intestine herniates into the scrotum or labia majora.
- c) In testicular feminization syndrome, uterus and uterine tubes are generally absent.
- d) Bilateral cryptorchidism results in sterility.
- e) None of the above.

_____ **68.** With regard to the **development of gonads**, which statement is **false?**

- a) Sex of an embryo is determined genetically at the time of fertilization.
- b) Germ cells do not appear in the genital ridges until the 6th week of development.
- c) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.
- d) The surface epithelium of the female gonad develop into oogonia.
- e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

_____ **71.** During the **development of the testes**, which statement is **false?**

- a) A dense tunica albuginea is the characteristic feature of testicular development in the fetus.
- b) Interstitial cells (of Leydig) begin to secrete androgenic hormones by the end of the third trimester.
- c) The fetal testis produces antimüllerian hormone by sertoli cells at 6-7 weeks
- d) Sertoli cells are derived from the surface epithelium of the testis.
- e) None of the above.

_____ **69.** With regard to the **vestigial structures derived from embryonic genital ducts**, which statement is **false?**

- a) The cranial end of mesonephric duct in males may persist as paradidymis.
- b) The epoophoron in the females correspond to the efferent ductules and duct of epididymis
- c) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
- d) The prostatic utricle is homologous to the vagina.
- e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

_____ **70. Concerning the development of the genital system, which statement is false?**

- a) Inguinal canals develop in both sexes.
- b) Sertoli cells constitute most of the seminiferous epithelium in the fetal testis.
- c) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.
- d) Gubernaculum does not play any role in the descent of ovaries.
- e) The persistent processus vaginalis in the female is called canal of nuck.

_____ **72. With regard to development of the ovaries, which statement is false?**

- a) The cranial part of the gubernaculum becomes the ovarian ligament.
- b) As the ovary separates from the regressing mesonephros, it is suspended by mesovarium.
- c) The surface epithelium becomes separated from the follicles in the cortex by tunica albuginea.
- d) The primordial follicles are derived from the rete ovarii.
- e) No oogonia are formed postnatally.

_____ **73. With regard to development of the eye, which statement is true?**

- a. Failure of choroid fissure to close results in congenital aphakia.
- b. The epithelium of the ciliary body is derived from endoderm.
- c. The optic nerve is a tract of diencephalon that is completely myelinated during the third trimester.
- d. The hyaloid canal (cloquet's canal) is found in the vitreous body.
- e. Aqueous humor is produced by the choroid plexus.

_____ **74. With regard to development of the ear, which statement is true?**

- a. The spiral organ of Corti is derived from neural crest.
- b. The middle ear cavity develops from the otic vesicle.
- c. The utricular portion of the otic vesicle gives rise to cochlear duct.
- d. Perilymph enters the subarachnoid space via the endolymphatic sac.
- e. None of the above.

_____ **76. Abnormal development of canal of Schlemm or the iridocorneal filtration angle causes**

- a. coloboma iridis
- b. congenital cataract
- c. congenital glaucoma

d. cyclopia

e. microphthalmia

_____ **78.** With regard to the development of the head and neck,

- a) sphenoidal sinuses develop during the early fetal life (10 to 11 weeks)
- b) the critical period of palate development is from the end of the 6th week until the beginning of the 9th week.
- c) taste buds form on the dorsal surface of the tongue only.
- d) all parenchymal (secretory) tissue of the salivary glands arises by proliferation of the oral epithelium.
- e) by late fetal period, the nasolacrimal duct drains into the inferior meatus in the lateral wall of the nasal cavity.

_____ **79.** Concerning the development of vasculature,

- a) distal parts of the 3rd pair of aortic arches join the dorsal aortae to form the internal carotid arteries.
- b) the left 4th aortic arch becomes the proximal part of the left subclavian artery.
- c) the distal part of the right 6th aortic arch becomes the ductus arteriosus.
- d) the superior vena cava forms from the right anterior cardinal vein and the right posterior cardinal vein.
- e) after birth, the proximal parts of the umbilical arteries persist as the superior vesical arteries.

_____ **77.** Utricular portion of the otic vesicle does not give rise to

- a. utricle
- b. semicircular ducts
- c. vestibular ganglion of CNVII
- d. endolymphatic duct and sac
- e. cochlear duct [organ of Corti]

_____ **80.** Development of the Integumentary system entails that,

- a) the epidermis is derived from surface ectoderm.
- b) castoff cells from the epidermis mix with secretions of sebaceous glands to form the vernix caseosa.
- c) teeth develop from ectoderm, mesoderm and neural crest cells.
- d) eccrine sweat glands develop from mesenchyme, which is derived from the mesoderm underlying the surface ectoderm.
- e) toenails and fingernails begin to develop at the tips of the digits postnatally.

_____ **81. With regard to the development of the digestive system,**

- a) the foregut may give rise to the lower respiratory system.
- b) the midgut may give rise to the descending colon and sigmoid colon.
- c) the hindgut derivatives are supplied by inferior mesenteric artery.
- d) during the rotation of the stomach, the lesser curvature moves to the left and the greater curvature moves to the right.
- e) the dorsal pancreatic bud forms the uncinate process and part of the head of the pancreas.

_____ **82. During the development of the digestive system,**

- a) omphalocele is a persistence of the herniation of abdominal contents into the proximal part of the umbilical cord.
- b) the superior two-thirds of the adult anal canal are derived from proctodeum.
- c) the junction of the epithelium is derived from the ectoderm of the proctodeum.
- d) the inferior one-third of the anal canal is supplied with superior rectal artery.
- e) the anal membrane normally breaks down by the end of the second trimester.

_____ **83. Concerning the development of vasculature,**

- a) distal parts of the 3rd pair of aortic arches join the dorsal aortae to form the internal carotid arteries.*
- b) the left 4th aortic arch becomes the proximal part of the left subclavian artery.*
- c) the distal part of the right 6th aortic arch becomes the ductus arteriosus.*
- d) the superior vena cava forms from the right anterior cardinal vein and the right posterior cardinal vein.*
- e) after birth, the proximal parts of the umbilical arteries persist as the superior vesical arteries.*

GOOD LUCK

37. Action potentials travel fastest along unmyelinated axons.fff
38. Action potentials normally travel from the axon terminal toward the cell body.fff
39. Unmyelinated axons conduct more slowly than myelinated axons. ttt
40. Unmyelinated axons generally have a smaller diameter than myelinated axons. ttt
41. Meissner's and Pacinian corpuscles are usually served by myelinated axons. Ttt
42. Dendrites are found in gray matter.tttt
43. Axons from receptors for slow, burning pain are typically small and unmyelinated.
44. White matter in the central nervous system consists of axon tracts with abundant myelin, where neuron cell bodies and dendrites are scarce or absent.
45. A single Schwann cell forms myelin around one and only one axon, while a single oligodendroglial cell forms myelin around several separate axons.
46. Myelinated axons found in white matter usually begin in gray matter and terminate in gray matter.
82. A typical peripheral mixed nerve includes all of the following EXCEPT:
- a. motor axons.
 - b. connective tissue of the perineurium and/or endoneurium.
 - c. Schwann cells.
 - d. sensory axons.
 - e. astrocytes.

Which the of the following cell types lines the central canal of the spinal cord and the ventricular system of the brain?

- a. astroglia
- b. oligodendroglia
- c. microglia
- d. Schwann cells
- e. ependymal cells

Choroid plexus consists of:

- a. dense regular connective tissue.
- b. white matter.
- c. gray matter.
- d. ependymal tissue and associated capillaries.
- e. arachnoid granulations.

An imbalance between production and drainage of cerebrospinal fluid can cause:

- a. Moniere's disease.
- b. astigmatism.
- c. presbyopia.
- d. glaucoma.
- e. hydrocephalus.

Pyramidal cells (neurons whose cell bodies have a characteristic, pyramidal shape) are characteristic of the:

- a. retina.
- b. cerebral cortex.
- c. cerebellar cortex.
- d. ventral horn of spinal cord.
- e. dorsal horn of spinal cord.

The electrical signal of an action potential is conveyed into the interior of each muscle fiber by:

- a. sarcomeres.
- b. transverse tubules.
- c. sarcoplasmic reticulum
- d. Z-lines.
- e. thick filaments.

Z-lines represent attachment sites for:

- a. thick filaments (myosin)
- b. thin filaments (actin)
- c. t-tubules
- d. sarcoplasmic reticulum
- e. neuromuscular junctions

Bone formation without a pre-existing cartilage model is called:

- a. intramembranous bone formation.
- b. extramembraneous bone formation.
- c. endochondral bone formation.
- d. trabecular bone formation.
- e. cancellous bone formation.

The activity of osteoclasts is inhibited, decreasing blood calcium levels, by secretion from the:

- a. thyroid follicles.
- b. adrenal cortex.
- c. parathyroid.

- d. thyroid parafollicular cells (C cells).
- e. neurohypophysis.

In the restructuring of the skeleton, bone remodeling:

- a. follows a cycle in which bone formation alternates with resorption.
- b. occurs exclusively in adult bone.
- c. occurs independently of soft tissue growth and external forces.
- d. only occurs at the epiphyses of long bones.
- e. results in the formation of cement lines.

In compact bone, osteocytes which are not adjacent to capillaries receive or exchange nutrients and metabolites by means of:

- a. processes contacting other osteocytes along canaliculi.
- b. lymph drainage through central (Haversian) canals.
- c. diffusion through the bony matrix.
- d. fenestrations in medullary vessels.
- e. active transport across the endosteum.

Substantia propria is the name for which tissue component of the cornea?

- a. epithelial tissue
- b. connective tissue
- c. nervous tissue.
- d. muscle tissue
- e. non-keratinized stratified squamous epithelium.

The iris is composed of several tissue types. Most of the visible portion is:

- a. adipose connective tissue.
- b. loose connective tissue.
- c. dense fibrous connective tissue.
- d. nervous tissue.
- e. epithelial tissue.

Which of the following structures is/are acidophilic?

- a. nucleoli.
- b. heterochromatin.
- c. fat droplets.
- d. ribosomes.
- e. mitochondria.

Continuous endothelium is found in:

- a. endocrine organs.
- b. kidney.

- c. lung.
- d. spleen.
- e. intestinal villi.

Lymphatic nodules are NOT found in:

- a. cortex of lymph nodes.
- b. tonsils.
- c. spleen.
- d. respiratory and gastrointestinal mucosa.
- e. thymus.

EMBRYOLOGY

- _____ 1. Which one of the following structures changes its position because of the **head fold**?
- a) Oropharyngeal membrane b) Septum transversum c) Primitive heart
d) a & b e) All of the above
- _____ 2. Most **birth defects** occur during:
- a) The first 2 weeks b) The embryonic period c) The fetal period
d) Early childhood e) None of the above
- _____ 3. The **cranial half of the 1st cervical sclerotome** gives rise to:
- a) Upper part of body of the 1st cervical vertebra. b) Lower part of body of the 1st cervical vertebra.
c) Whole body of the 1st cervical vertebra. d) Part of base of occipital bone.
e) Part of roof of the cranial cavity
- _____ 4. Which of the following **muscles** develop from the epimeres?
- a) Intercostal muscles. b) The strap muscles of the neck. c) Rectus abdominis.
d) Quadratus lumborum. e) Extensor muscles of the back.
- _____ 5. Failure of **fusion of the neural arches** of the developing vertebrae could lead to:
- a) Achondroplasia b) Acromegally c) Spina bifida d) Brachycephaly e) Amelia
- _____ 6. Which of the following gives rise to most muscles of the tongue?
- a) Lateral mesoderm b) Local splanchnic mesenchyme
c) Myotomes of the occipital somites d) All of the above e) None of the above
- _____ 7. Frontonasal prominence gives rise to the:
- a) Forehead b) Sides of the face c) Sides of the nose d) All of the above e) None of the above
- _____ 8. Which of the following is **not** a component of vernix caseosa?
- a) Secretions from sebaceous gland b) Secretions of apocrine sweat glands
c) Fallen epidermal cells d) Fallen hairs e) None of the above
- _____ 9. Structures derived from endoderm of the second pharyngeal pouch include:

- a) Thymus gland
- d) Parafollicular cells of thyroid gland

- b) Palatine tonsil
- e) All of the above

- c) Thyroid gland

_____ 10. Germ layer that does **not** give rise to the tympanic membrane

- a) Ectoderm
- b) Mesoderm
- c) Endoderm
- d) b & c
- e) None of the above

_____ 11. The muscular part of interventricular septum is formed by:

- a) Downward extension of the spiral aorticopulmonary septum
- b) Fusion of superior and inferior endocardial cushions of the ventricular wall
- c) Fusion of right and left endocardial cushions of the ventricular wall
- d) Upward growth of a midline septum from the floor of a common ventricle
- e) Expansion of the lateral aspects of primitive ventricle with the midline region remained unexpanded

_____ 12. The common carotid arteries develop from:

- a) Truncus arteriosus
- b) 1st aortic arch arteries
- c) 2nd aortic arch arteries
- d) 3rd aortic arch arteries
- e) Dorsal aortae

_____ 13. In fetal blood circulation oxygenated blood is not contaminated by deoxygenated blood in which of the following sites?

- a) Left atrium
- b) Right atrium
- c) Inferior vena cava
- d) Descending aorta
- e) None of the above

_____ 14. Which of the following veins participates in the formation of the inferior vena cava?

- a) Right vitelline vein
- b) Right posterior cardinal vein
- c) Right umbilical vein
- d) Left umbilical vein
- e) Right posterior cardinal vein

_____ 15. Portal vein is derivative of:

- a) Right vitelline vein
- b) Left vitelline vein
- c) Anastomosis between left and right vitelline veins
- d) Anastomosis between right umbilical and right vitelline veins
- e) Anastomosis between left umbilical and left vitelline veins

_____ 16. After birth umbilical vein becomes:

- a) Medial umbilical ligament
- b) Median umbilical ligament
- c) Lateral umbilical ligament
- d) Ligamentum teres hepatis
- e) Ligamentum venosum

_____ 17. Which of the following cells of the lung develop from mesoderm?

- a) Type I Pneumocytes.
- b) Type II Pneumocytes.
- c) Alveolar dust cells.
- d) Clara cells
- e) Goblet cells

_____ 18. Which of the following statements is correct regarding to the development of the gut?

- a) Esophagus was short during early development, but gets elongated towards the end of embryonic period.
- b) The midgut was connected to the yolk sac by the allantois.
- c) The primitive gut tube is in open communication with the amniotic cavity.
- d) The site of junction of foregut and midgut derivatives is the 2nd part of duodenum just proximal to the entrance of common bile duct.

e) All are correct

_____ **19.** Concerning the development of pancreas

- a) The ventral bud arises from the proximal part of haptic bud.
- b) The ventral bud rotates & comes to lie anterior to the dorsal bud.
- c) The ventral bud arises from the distal part of foregut, while the dorsal bud arises from the proximal part of midgut.
- d) The exocrine portion develops from endoderm, while the endocrine portion develops from local mesoderm.
- e) The ventral bud appears before the appearance of the dorsal bud.

_____ **20.** Which of the following statements is **not correct** regarding to the development of the coelomic cavities?

- a) Peritoneal cavity was temporarily divided into two by the gut and its primitive mesenteries.
- b) Develop within the intermediate mesoderm.
- c) Are lined by both splanchnic and somatic mesoderm layers.
- d) Primitive pericardial cavity gives rise to the serous pericardium.
- e) Pericardioperitoneal canals are invaginated by the growing lungs.

_____ **21.** The crura of the diaphragm develops from:

- a) Septum transversum.
- b) Dorsal mesentery of the esophagus.
- c) Pleuroperitoneal membrane.
- d) Myotomes of the body wall.
- e) None of the above.

_____ **22.** Which one of the following is **not** a derivative of the neural crest?

- a) Adrenal chromaffin cells
- b) Melanocytes
- c) Odontoblasts
- d) Enteric nervous system
- e) None of the above

_____ **23.** Which of the following nuclei is **not** developed from the alar plate of the neural tube?

- a) Olivary nucleus
- b) Cuneate nucleus
- c) Thalamus
- d) Superior colliculus
- e) None of the above

_____ **24.** Primary brain vesicle that remains relatively unchanged as compared to the others is:

- a) Rhombencephalon
- b) Prosencephalon
- c) Telencephalon
- d) Mesencephalon
- e) Diencephalons

_____ **25.** During the development of the central nervous system zone of the wall of the neural tube to be occupied by neuroblasts forming the gray matter is:

- a) Marginal zone
- b) Ventricular zone
- c) Mantle zone
- d) Ependymal zone
- e) Neuroepithelial zone